

# Aayush Patel

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*Applied AI engineer at Georgia Tech — production RAG systems, scalable end-to-end ML pipelines, feature engineering*

## EDUCATION

### Georgia Institute of Technology — Atlanta, GA

Expected May 2028

*B.S. Computer Science — Intelligence & Systems Architecture / Minor: FinTech / GPA: 3.71 / 4.00*

**Coursework:** Machine Learning, Artificial Intelligence, Advanced Data Structures & Algorithms, Objects & Design

## TECHNICAL SKILLS

**ML & AI Systems:** PyTorch, scikit-learn, Pandas, NumPy, Retrieval-Augmented Generation (RAG), Prompt Engineering, Structured Extraction, LLM API Integration (OpenAI, Gemini, Claude, Vertex AI), Vector Search (pgvector/HNSW, ChromaDB), LSTM, ARIMA, Random Forest, XGBoost, CNN, Evolutionary Algorithms (deap), NLP

**Languages:** Python, SQL, Java, C, C++, JavaScript, TypeScript

**Infrastructure & Tools:** FastAPI, Flask, PostgreSQL, pgvector, Docker, Docker Compose, CI/CD (GitHub Actions), Google Cloud Platform (GCP), REST APIs, Next.js, Firebase/Firestore, Pydantic, WebSockets, Git

## WORK EXPERIENCE

### Georgia Institute of Technology

May 2026 – Aug 2026

*Teaching Assistant, CS 2340: Objects & Design — Atlanta, GA*

- Developed OOP, design-pattern, and Agile/Scrum materials for 200+ students across CS 2340 sections
- Facilitated office hours and technical presentations on project architecture, debugging techniques, and iterative code review
- Recognized by faculty for documenting software design trade-offs to make it clear to students of varied backgrounds

### Big Data Big Impact, Georgia Tech

Jan 2026 – Present

*Director of Projects — Atlanta, GA*

- Led 11 AI/ML project teams (100+ students) across NLP, forecasting, and computer-vision domains
- Set uniform model-evaluation standards: precision/recall dashboards, confusion matrices, stratified CV, AUC score
- Drove 6 projects to deployment; presented results to College of Computing Advisory board and a Microsoft industry panel

### Georgia Tech VIP — EMADE Lab (Advisor: Dr. Jason Zutty)

Aug 2025 – May 2026

*Undergraduate Researcher — Evolutionary AutoML & LLM-Guided Synthesis — Atlanta, GA*

- Designed LLM-as-mutation-operator framework; integrated Gemma, Qwen, Gemini, and DeepSeek-R1 in *deap* framework
- Improved synthesis solution quality by 14% on TSP and symbolic regression over human-engineered baselines
- Engineered island-migration scheduler for parallel evolution; achieved 5.8× convergence speedup

### Mascot Forge Pvt Ltd (automotive-parts manufacturer, India)

May 2025 – Aug 2025

*Data Engineering Intern*

- Built gradient-boosted demand-forecasting model with scikit-learn on production records requiring rigorous pre-processing
- Validated on held-out data; model adopted by operations management, cutting reporting latency by 1.3 hours
- Built PostgreSQL analytics dashboards; translated model findings into executive-level ROI presentations

## PROJECTS

### Research Paper RAG Platform

*FastAPI / pgvector / Next.js / Docker / Claude API*

- Built production RAG pipeline: FastAPI ingestion, page-aware PDF chunking, pgvector/HNSW semantic index
- Implemented Pydantic-validated extraction with field-level citations and weak-evidence flagging
- Evaluated retrieval quality via Recall@k and MRR; containerized full stack with Docker Compose

### Vitalis — AI Health Assistant

*Gemini API / ChromaDB / FastAPI / JSON Schema*

- Built an agentic Gemini tool-use pipeline for multi-step document parsing, retrieval, and synthesis
- Integrated ChromaDB as persistent agent memory; retrieved context via cosine similarity for structured prompts
- Applied JSON schema constraints on agent outputs; improved structured retrieval precision by 31%

### TerraTrends — County-Level Economic Forecasting

*Python / LSTM/ARIMA / scikit-learn / Flask / deck.gl / GCP*

- Trained LSTM and ARIMA ensemble on multi-source economic features; validated on held-out county data splits
- Automated ETL from 15+ public economic datasets on GCP; surfaced model predictions via deck.gl dashboard

### Flight Delay Prediction

*Python / scikit-learn / XGBoost / Flask / Pandas*

- Trained CNN, RF, XGBoost, and MLP ensemble on 6M+ flight-weather records; achieved 92% accuracy and AUC 0.94
- Applied RFE on weather-derived features; improved precision by 14%; deployed predictions via Dockerized Flask API